

ECE 315

Steven Knudsen, PhD, PEng

Lecture 11 :

- Computer Interfacing
- Overview of some interfaces
- Duplex
- The nature of analogue and digital signals



**UNIVERSITY
OF ALBERTA**

Last Lecture

- Mutexes and Semaphore example
- Synchronization
- Priority Inversion
- Deadlock

This Lecture

- Computer Interfacing
- Overview of some interfaces
- Duplex
- The nature of analogue and digital signals

Learning Objectives

- Understand that to be useful computers must interface with the outside world
- Recognize a few common computer interfaces, what is connected to the computer
- Know the difference between simplex, half- and full- duplex
- Appreciate that all signals are analogue and some of what that means in terms of using them for interfacing computers
- Be able to say how DC offsets and noise might affect a signal
- Describe a ground loop and its effects

What is Computer Interfacing?

- In this course we look at the low-level aspects of computer interfacing; i.e. interfacing of embedded computers
- It is very similar to what is done at higher-levels, non-embedded computing systems, except that at that level there are more utilities, more computational resources, and so on to help
- Thus, computer interfacing is about connecting embedded systems to external devices (sensors, actuators, computers).
- Its purpose is to enable data exchange, control, and monitoring

Which Interfaces?

- **Serial Interfaces:** UART, SPI, I²C — short-range, low data rate.
- **Parallel Interfaces:** Faster but use more pins (e.g., LCD data buses).
- **Memory interfaces:** Serial and Parallel, direct memory access (DMA)
- **Wireless Interfaces:** Bluetooth, Wi-Fi, Zigbee.

Serial Interfaces

- A **serial interface** is a communication mechanism that transfers data one bit at a time over a single or a small set of communication lines, typically between a microcontroller and a peripheral device such as a sensor, display, or another microcontroller

Key Characteristics

- **Bit-by-bit transmission:** Data is sent sequentially rather than in parallel (all bits at once), which reduces the number of physical connections required.
- **Clock synchronization:** **Synchronous** interfaces use a shared clock signal, while **asynchronous** interfaces rely on timing agreements between devices
- **Directionality:** Can be **unidirectional** (one-way) or **bidirectional** (two-way communication)

Common Serial Interfaces in Embedded Systems

- UART (Universal Asynchronous Receiver/Transmitter):
 - Asynchronous communication (no shared clock).
 - Typically used for debugging or communication between controllers and PCs.
- SPI (Serial Peripheral Interface):
 - Synchronous, *full-duplex*.
 - Often used for fast communication with sensors or memory chips.

Common Serial Interfaces in Embedded Systems

- I²C (Inter-Integrated Circuit):
 - Synchronous, *half-duplex*.
 - Supports multiple masters and slaves on the same bus.
- CAN (Controller Area Network):
 - Used in automotive and industrial systems.
 - Allows multiple microcontrollers to communicate without a central host.

What's all this Duplex Stuff, Anyhow?

- Refers to the direction and concurrency of data transmission
- There are three common modes
 - **Simplex** – data travels in only one direction, ever
 - **Half-Duplex** – data travels in two directions, but never at the same time
 - **Full-Duplex** – data travels in two direction simultaneously

Simplex



Advantages

- Simplicity: simple to implement; data in only one direction
- Cost-Effective: hardware required (e.g., cables, connectors) can be less costly
- No Collision : data in only one direction, no risk of data collision; communication is secure and consistent
- Efficient Use for Specific Applications: Ideal for applications where only one-way communication is necessary, such as broadcasting, keyboard input to a computer, or sending data to a printer



Made with help from ChatGPT

Disadvantages

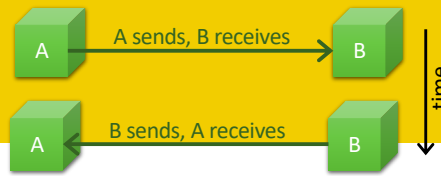
- Lack of Bidirectional Communication: The most significant disadvantage is the incapacity to send data back in the opposite way. This limitation makes it unsuitable for interactive communication.
- Inefficient for Complex Tasks: not suitable for tasks requiring response or acknowledgment
- Limited Flexibility: cannot be easily adjusted to situations needing bidirectional comms
- Not Ideal for Modern Networks: Most advanced communication systems require bidirectional data flow, making simplex mode largely antiquated in networking scenarios

11

Broadcast communications is an example where simplex is best, the only way to make it work by definition.

However, broadcast is dying out because even though it's more complex asymmetric full-duplex has significant advantages.

Half-Duplex



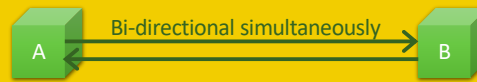
Advantages

- Efficient Use of Channel: bidirectional communication over a **single channel**
- Cost-Effective: needs less complicated and less costly hardware than full-duplex systems
- Simplified Collision Handling: only one device can transmit at a time, complicated collision detection and management protocols not needed
- Suitable for Periodic Communication: Optimal when communication doesn't need to happen concurrently in both directions, such as walkie-talkies or specific network protocols.

Disadvantages

- Slower Data Transmission: data can only run in one direction at a time; communication is delayed compared to full-duplex systems
- Increased Latency: The need for nodes to take turns introduces delay, which can affect performance in congestion applications
- Not Ideal for High-Traffic Networks: In networks with congestion or requiring constant communication, half-duplex mode can become a restriction
- Inefficient for Complex Communications: When frequent two-way communication needed, it's less effective as it forces devices to wait their turn, which can lead to pauses

Full-Duplex



Advantages

- High-Speed Communication: Data transfer is quicker because there is no delaying for the channel to clear before sending or receiving
- Reduced Latency: Communication is parallel, time lag is minimized, which is critical for real-time applications like web conferencing
- Better Utilization of Bandwidth: The maximum available transmission capacity is available

Disadvantages

- Complexity and Cost: Requires more complicated hardware and can be more costly than simplex or half-duplex systems.
- Requires Quality Infrastructure: Demands an advanced quality of architecture, such as better cabling and more refined networking equipment, to avoid interference and assure smooth data flow.

In reality, the hardware costs for full-duplex systems is falling allowing it's use in more situations. The convenience, low-latency, high-throughput make it very attractive.

The Nature of Signals in Interfaces

- Like to think of our world as being digital...
- Real world is always analogue, so we must deal with the nature and limitations of analogue signals before we think of digital data
- The aim is to detect information that is present in the analogue domain.
- To do this, we implement hardware and software to process the signals.

Analogue Signal Management and Processing

1) Analog signal buffering, conditioning and filtering

- Use buffers/drivers to amplify signals & reduce output impedance
- Noise rejection filters, anti-aliasing filters prior to analogue-to-digital conversion (A/D or ADC)

2) Impedance matching to avoid unwanted signal “ringing” (oscillations)

- Avoid sudden changes in characteristic impedance of connections
- Use a **parallel termination** resistance at the far end of a wire
- Use a **series termination** resistance at the near end of a wire

Analogue Signal Management and Processing

3) Analog-to-digital (ADC) and digital-to-analog signal conversion (DAC)

- **ADC** – Analog inputs from sensors need to be filtered, sampled and converted into digital form to permit digital signal processing (DSP)
- **DAC** – Digital outputs need to be converted into voltage or current signals to control actuators, heaters, light sources, radio signals, etc.

4) Digital signal level conversion and signal isolation

- Voltage level conversion (3v3 → 5v), voltage-current conversion
- Optical isolation transceivers, isolation transformers

Digital Signal Management and Processing

5) Digital modulation of an analog signal

- Digital signals must be encoded as analog signals of controlled bandwidth to propagate well through the passband of a communication channel
- *Modem* = *modulator* (in transmitter) + *demodulator* (in receiver)

6) Timing recovery, demodulation & decoding a modulated signal.

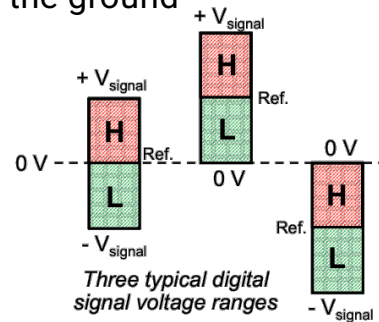
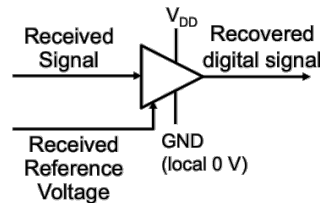
- Steps that are present in many digital communication receivers.
- Error-correcting codes are used to detect and correct digital errors.

Digital Signal Management and Processing

- 7) Synchronization, framing, handshaking, error detection
 - Start (for *synchronization*) and stop bits (for *spacing*) in RS-232C
 - Request-to-send and clear-to-send handshaking (for flow control)
 - Parity bit (or checksum) generation at the transmitter
 - Parity bit (or checksum) re-generation & checking at the receiver.
 - *One parity bit* permits the *detection of an odd number of bit errors*.
- 8) Data buffering and flow control to handle rate fluctuations
 - Temporary storage in buffers inside transmitters and receivers
 - A *buffer accommodates short-term data rate fluctuations*, but *flow control* is required to *handle longer-term mismatches in data rates*.

Unbalanced (or Single-ended) Digital Signalling

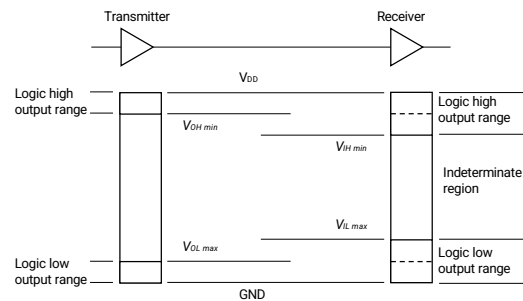
- A single-ended signal comprises two wires, one that carries the signal, the other that is the reference for the signal
 - Usually, the second wire is set to or defined as ground
- **Advantages:** The reference wire can be shared for multiple signals.
- **Disadvantages:** Noise added to the signal and/or the ground reference (e.g., ground loop noise) can cause bit errors at the receiver



Noise Margin in Digital Signaling

- Noise Margin is the maximum noise voltage that a digital circuit can tolerate without misinterpreting a logic level.
- It measures the robustness of digital signals against electrical noise and voltage fluctuations.

Parameter	Description
V_{OH}	Min output voltage recognized as logic 1
V_{OL}	Max output voltage recognized as logic 0
V_{IH}	Min input voltage guaranteed to be logic 1
V_{IL}	Max input voltage guaranteed to be logic 0
NM_H	Noise Margin for high signal
NM_L	Noise Margin for low signal



$$NM_H = |V_{OHmin} - V_{IHmin}|$$

$$NM_L = |V_{ILmax} - V_{OLmax}|$$

Made with help from ChatGPT

20

<https://www.sciencedirect.com/topics/computer-science/noise-margin>

Noise Margin in Digital Signaling

Technology	Noise-Margin Measures (V, volts)					
	V_{DD}	V_{OH}	V_{IH}	V_{TH}	V_{IL}	V_{OL}
5 V CMOS	5	5.0	4.44	2.5	1.5	0.5
5 V TTL	5	5.0	2.4	1.5	0.8	0.4
3.3 V LVTTTL	3.3	3.3	2.4	1.5	0.8	0.4
2.5 V CMOS	2.5	2.5	2.0	1.2	0.7	0.4
1.8 V CMOS	1.8	1.8	1.45	0.9	0.65	0.45

Balanced (or Differential) Digital Signaling

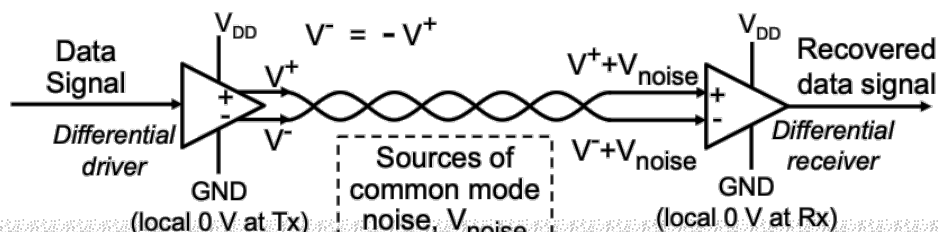
- Two wires – **no ground (GND) is needed**
- The transmitter sends two signals; V^+ and the inverse signal, V^-
$$V^- = -V^+$$
- The receiver subtracts the two to recover the single-ended signal
- If V^+ and V^- are corrupted by common-mode additive noise, V_{noise} , then the noise is cancelled out by subtraction:

$$(V^+ + V_{noise}) - (-V^+ + V_{noise}) = V^+ - (-V^+) = 2 V^+$$

Advantage: Common-mode noise is cancelled at a differential receiver.

Balanced (or Differential) Digital Signaling

- Typical sources of additive noise: nearby signals, changing electric & magnetic fields produced by power supplies, motors, transformers, etc.
- The noise rejection is even more effective if the wires are twisted because this increases the chance that noise is more equally induced on both wires of a pair.



Unwanted Signals Across Systems

- Noise, DC offsets, cross-talk (strong signals are coupled into circuits) can wreak havoc with digital (and audio systems)
- **DC Offset:** A constant voltage ($\pm$) added to a signal, shifting its baseline away from zero.
 - Caused by ground loops, sensor bias, or mismatched circuits.
 - Leads to distortion, reduced dynamic range, and inaccurate measurements.
- **Noise:** Random, unwanted variations superimposed on a signal.
 - Sources include power supply ripple, electromagnetic interference (EMI), and digital switching.
 - Can obscure weak signals and reduce signal-to-noise ratio (SNR).

Effects of DC Offsets and Noise

- **Measurement Errors:** DC offsets bias analog-to-digital converters (ADCs), leading to incorrect readings.
- **Signal Distortion:** Noise and offsets alter waveform integrity, especially in audio and sensor systems.
- **Ground Loop Problems:** Differences in ground potential introduce hum or DC currents through shared grounds.
- **System Instability:** Feedback systems can behave unpredictably if offsets or noise corrupt reference signals.

Effects of Noise and Cross-talk

Noisy SPI **chip**
select signal



26

From a student project that was fixed and ended up going to space and doing very well.

Effects of Noise and Cross-talk

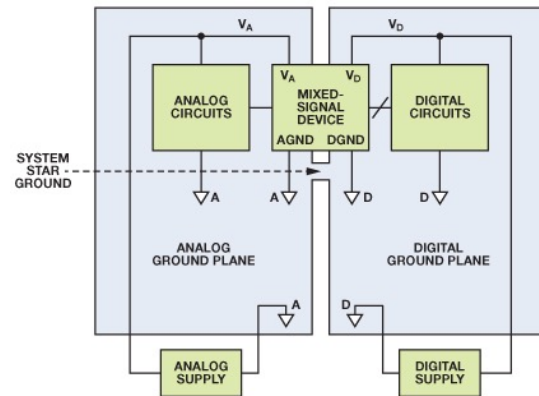
Noisy SPI **chip**
select signal and
data signal



The green is the data signal is very noisy, which was a result of having the analogue and digital grounds all mixed together.

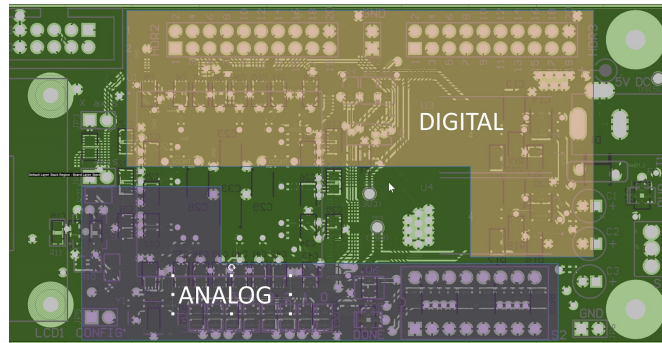
Noise and Unwanted Signals Across Ground Planes

- This is a circuit-level problem and a result of mixing analogue and digital grounds
- The solution is to have all the digital circuits ground to a common plane, all the analogue circuits ground to a different common plane, and have the two plans connected at only one point
 - Sometimes called a “star ground”



Noise and Unwanted Signals Across Ground Planes

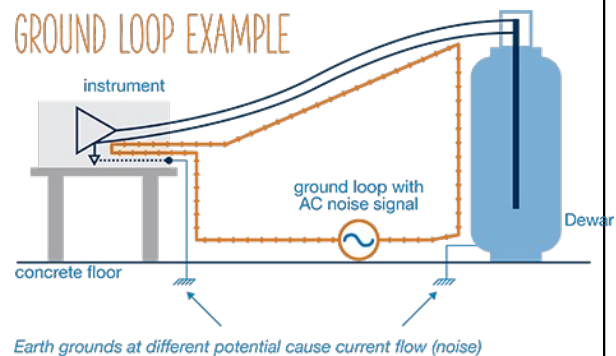
- The common point does not have to be small, restricted
- Careful layout of circuits can achieve the same result
 - The “star” connection is just bigger



<https://resources.altium.com/p/how-to-use-a-star-point-for-analog-ground-digital-ground-connection>

Ground Loops

- Systems that are physically far apart may have grounds that are at different potentials with respect to each other.
- When connected by a wire, a current will flow and signals at either end will have different references, i.e., there will be a DC offset
- Chassis grounds are at different potentials



<https://blog.lakeshore.com/avoiding-ground-loops-in-electronic-device-measurements> 30

These systems are not really all that far apart, yet there can be a ground potential difference.

I have seen this in a capstone lab between work benches.

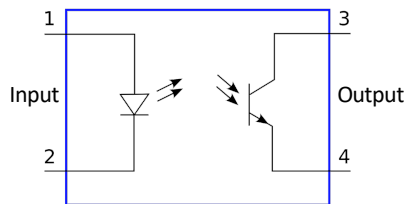
That case may have been due to poor wiring, there being bad connectors that result in relatively high resistance. Then, according to Ohm's Law, we will see a voltage across that resistance that will make the supposed 0V reference non-zero.

Electrical Isolation Between Systems

Goal: Break electrical connection to eliminate DC paths while maintaining signal transmission.

Opto-Isolation (Optocouplers):

- Converts electrical signal to light and back to electricity.
- Provides galvanic isolation, preventing DC offset transfer.
- Common in digital signal transmission and control circuits.

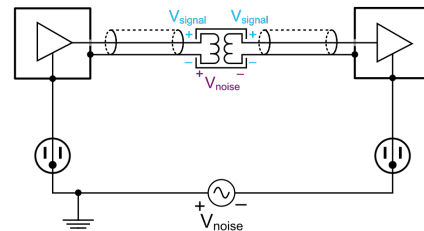


UNIVERSITY
OF ALBERTA

By Riflemann~commonswiki - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=1545090>

Transformer Isolation:

- Transfers energy via magnetic coupling.
- Blocks DC and low-frequency noise.
- Used in power supplies and analog signal paths.



Hddharvey, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>,
via Wikimedia Commons

Made with help from ChatGPT 31

The solution is to break that current loop, which can be done using opto-isolation or transformer isolation.

Next Lecture

- Serial Interfacing
- UART
- RS-232C, RS-422, RS-485

Questions

